

E-Commerce Order Management System

Task – I: Requirement Analysis and Customer Database Module

1. Project Title

E-Commerce Order Management System

2. Objective

The main objective of the E-Commerce Order Management System is to manage customers, products, orders, payments, and deliveries efficiently. The system should reduce manual work, maintain accurate records, and provide quick access to order-related information.

3. Requirement Analysis

Functional Requirements

Customer Management

- Register new customers.
- Store customer details such as name, email, phone number, and address.
- Update and delete customer information.
- Search and view customer details.

Product Management

- Add, update, and remove products.
- Store product name, price, category, and stock quantity.
- Check product availability.

Order Management

- Customers can place orders.
- Generate a unique order ID.
- Store ordered products, quantity, price, and order date.
- Update order status such as Placed, Shipped, Delivered, or Cancelled.
- Allow customers to view their order history.

Payment Management

- Record payment details.
- Maintain payment status such as Pending, Paid, or Failed.
- Link payments with the corresponding order.

Delivery Management

- Store shipping and delivery information.
- Track delivery status.
- Update delivery details when required.

Admin Management

- Admin can manage customers, products, orders, payments, and deliveries.
- Admin can generate basic reports.

4. Non-Functional Requirements

Security: Customer and payment information must be protected.

Performance: The system should respond quickly to user requests.

Reliability: Data should be stored accurately without unnecessary loss.

Scalability: The system should support an increasing number of customers and orders.

Usability: The interface should be simple and easy to understand.

Maintainability: The system should be easy to modify and maintain.

5. Customer Database Module

The Customer Database Module is responsible for storing and managing customer information.

Field	Data Type	Constraint	Description
Customer_ID	INT	Primary Key	Unique customer ID
First_Name	VARCHAR(50)	NOT NULL	Customer first name
Last_Name	VARCHAR(50)	NOT NULL	Customer last name
Email	VARCHAR(100)	UNIQUE, NOT NULL	Customer email
Phone	VARCHAR(15)	UNIQUE	Contact number
Address	VARCHAR(255)	NOT NULL	Customer address
City	VARCHAR(50)	NOT NULL	Customer city
State	VARCHAR(50)	NOT NULL	Customer state
Pincode	VARCHAR(10)	NOT NULL	Postal code
Registration_Date	DATE	NOT NULL	Date of registration

6. Main Operations

- Create: Add a new customer.
- Read: View/search customer information.
- Update: Modify customer details.
- Delete: Remove a customer record.
- Search: Find customers using Customer ID, name, email, or phone number.

7. Expected Outcome

The system will provide a centralized database for managing customers and their orders. It will improve data accuracy, simplify order processing, and allow administrators to efficiently manage the e-commerce business.